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In situ jet energy calibration via γ-jet p_{T} balance in p+p collisions at $\sqrt{s}$=200 GeV with the sPHENIX detector

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And others...

Abstract

This sPHENIX note details the γ-jet study used for the *in situ* jet energy calibration using p+p collisions at $\sqrt{s}$=200 GeV taken in 2024. "*In situ*" in this context means using well-understood physics processes inside the same dataset to calibrate jets, i.e., here using a well calibrated direct photon as measured in the Electromagnetic Calorimeter (EMCal) that is balanced against the reconstructed anti-k_T jet measured in both the EMCal and Hadronic Calorimeters (HCal) with a range of transverse momenta and size R parameters. Resulting energy calibration values including systematic uncertainties are quoted in Section 6.

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1 Introduction

In situ jet energy calibration is a data-driven procedure used to determine residual corrections to the jet energy scale (JES) directly from collision events. After simulation-based calibrations account for detector response, pileup, and absolute energy scale effects, remaining mismodeling is removed by exploiting transverse momentum balance in well-understood physics processes. At the Large Hadron Collider, experiments such as ATLAS and CMS calibrate jets by comparing the reconstructed jet transverse momentum to that of precisely measured reference objects, thereby anchoring the jet response to objects with accurately known energy scales [1, 2].

The primary in situ methods rely on momentum balance in γ+jet and Z+jet events, where the photon or dilepton system provides a high-precision reference. In these topologies, the jet response is extracted from the ratio $x_J = p_T^{\text{jet}} / p_T^{\text{ref}}$, after correcting for additional radiation and resolution effects. Dijet events are used for η-intercalibration by balancing a forward jet against a well-calibrated central jet, while multijet balance techniques extend the calibration to very high transverse momentum [1, 2]. These complementary channels ensure coverage across detector regions and p_T ranges.

Systematic uncertainties arise from parton shower modeling, out-of-cone radiation, underlying event contributions, and the energy scale of the reference object. The resulting JES corrections and their uncertainty components are propagated to physics measurements, where they often constitute a leading systematic effect in precision measurements and searches for new phenomena [1, 2]. Continuous validation with data is therefore essential to maintain percent-level control of jet-related uncertainties.

For the sPHENIX experiment, the primary *in situ* channel is balanced γ+jet events. Figure 1 shows the projection from the sPHENIX 2020 Beam Use Proposal for the statistical uncertainty on the energy scale using the γ+jet channel as a function of jet p_{T}. In 2024, sPHENIX sampled approximately 107 pb^{-1} of calorimeter data for collisions over all z-vertices – as shown in Figure 2.

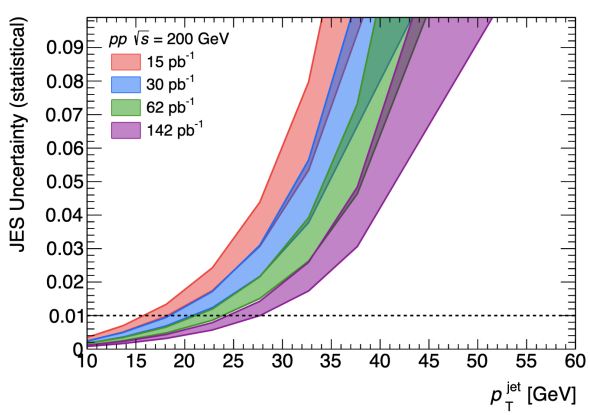

Figure 1: The projected sPHENIX statistical uncertainty contribution to the Jet Energy Scale (JES) uncertainty as determined from the "golden channel" via γ+jet direct balance studies. The dashed line at 1% represents a final JES uncertainty goal. This figure is reproduced from the sPHENIX 2020 Beam Use Proposal.

For this note, jets are reconstructed with the anti-k_T algorithm using the FASTJET package [5]

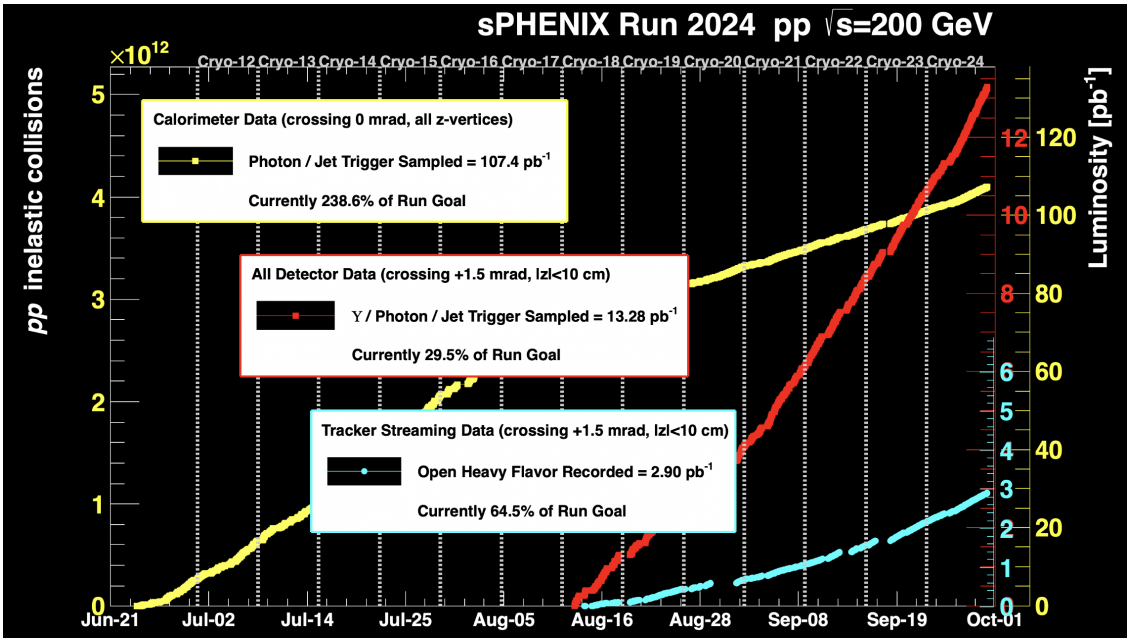

Figure 2: sPHENIX integrated luminosities from p+p data taking in 2024.

and studies with a set of radius parameters $R = 0.4, 0.6, 0.8$. The sPHENIX preliminary $R = 0.4$ jet cross section in p+p collisions, as documented in sPHENIX internal note [7], utilized a $\pm 8\%$ systematic uncertainty on the overall jet energy scale based on a combination of EMCal calibrations based on π^0 and η masses and widths and HCal calibrations from test beam. The goal of this analysis is set the overall jet energy scale *in situ* and thus to reduce the systematic uncertainty to the range 1-3%. These studies include examining any potential p_T or R dependence in the energy scale.

2 Data and Monte Carlo samples

For this analysis, photon candidates and jets are reconstructed using EMCal only and EMCal plus HCal tower energies, respectively, i.e., charged particle tracking is not considered in this analysis. In terms of work flow, data an MC DSTs are passed over and reconstructed photons and jets of radius $R = 0.4, 0.6$, and 0.8 are saved for selected events. In the Monte Carlo (MC) samples the identical information is saved, as well as truth-level photons and truth-level jets within those events. For output to ROOT **TTrees** [4], there are no selections on shower shape, timing, or the kinematic relationship between the photons and jets.

2.1 Data

The data samples used for this analysis are the entire golden run list (GRL) from p+p data taken in 2024. `https://github.com/sPHENIX-Collaboration/analysis/blob/master/runListGeneration/Run2024pp/Full_ppGoldenRunList_Version1.list`

The DST used for this analysis are the DST JETCALO DSTs, ana 521, cdb tag 2025p007_v001

This includes both data taken at 1.5 mrad crossing angle (run numbers 51093-53864) and at 0.0 mrad crossing angle (run numbers 47289-51015).

2.1.1 Event Selection

The only trigger selections used in this analysis were the MBDNS>= 1 && Photon > 5 GeV trigger for the 0 mrad runs and MBDNS>= 1 && Photon > 5 GeV && $|zvtx| < 10$ cm for the 1.5 mrad runs. The scaledown for this trigger was set to zero, i.e., record every live trigger, for the entirety of running. A reconstructed MBD z-vertex of $|z|$ <60 cm is applied.

Event information is only stored in the **TTrees** if an EMCal cluster, i.e., a photon candidate) with $p_{\mathrm{T}} > 10$ GeV is reconstructed and an uncalibrated jet of any radius with $p_{\mathrm{T}} > 3$ GeV is reconstructed in the event or a calibrated jet of any radius with $p_{\mathrm{T}} > 5$ GeV is reconstructed. If a jet passes either the uncalibrated or calibrated cut, both energy scales are saved.

2.2 Monte Carlo

The MC samples used for this analysis are generated with the PYTHIA-8 package [8]. In sPHENIX nomenclature, they are the samples Photon 5, 10, 20 GeV, and Jet 5, 10, 20, 30, 50, 70 GeV. The minimum bias sample is not used.

For the MC samples, the following DSTs and tags are used:

- DST_TRUTH_JET
- DST_MBD_EPD
- DST_CALO_CLUSTER
- G4Hits
- -run 28
- -nop

Because there is significant overlap between leading photon energy in the MC samples, there are cutoffs on both the truth and reconstructed p_{T} of the leading photon/cluster to ensure the combined sample is smooth. After that one may combine the samples by

1. Scale the yields by the number of events in the sample
2. Scale the yields by the relative cross sections
3. Add the histograms together.

The combining of the samples is done following the recommendation at `https://wiki.sphenix.bnl.gov/index.php?title=Jet_Structure_Topical_Group`.

As a systematic uncertainty check – see details in Section 5.3 – similar samples are being generated with the HERWIG generator [3].

3 Photon and jet selection

As detailed earlier, events are stored to **TTrees** if they have both a reconstructed cluster and jet above the given thresholds. Then the analysis passes over the **TTrees** and selections are made on the photon candidate and jet to determine if they are included in the analysis.

1. Find the leading p_{T} cluster passing a timing cut of 0 ns $< \Delta t_{MBD,cluster} <$ 4 ns in the event with no η requirement.
2. Find the leading p_{T} jet in the event that is greater than ΔR=0.4 from the cluster and passes a timing cut of 0 ns $< \Delta t_{MBD,jet} <$ 4 ns. There is also no eta cut
3. once the cluster and jet selections have been maid, the pair must pass the following cuts
   - $|\Delta t_{\mathrm{cluster,jet}}| < 4$ ns
   - $|\eta_{\mathrm{cluster}}| < 1.1$
   - $|\eta_{\mathrm{jet}}| < 1.1 - R_{\mathrm{jet}}$
   - finally, require the photon candidate and jet be back-to-back in azimuth via $\Delta\phi_{\gamma,\mathrm{jet}} > 7\pi/8$
4. If the pair passes all cuts, the cluster isolation energy and BDT score will determine which ABCD region it belongs to – see details in Section 5.1.

If the photon-jet pair fail any of the cuts, the event is rejected. Otherwise, their ratio is included in the relevant x_J plot.

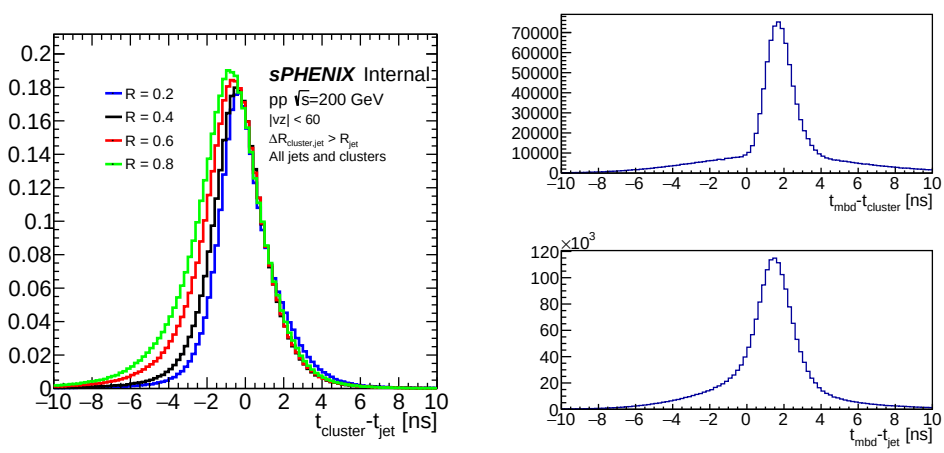

Figure 3: (Left) the EMCal cluster time minus the Jet time is shown for different jet size parameters (R). (Right) The time distribution from the MBD minus the time of the EMCal cluster (upper) or Jet (lower).

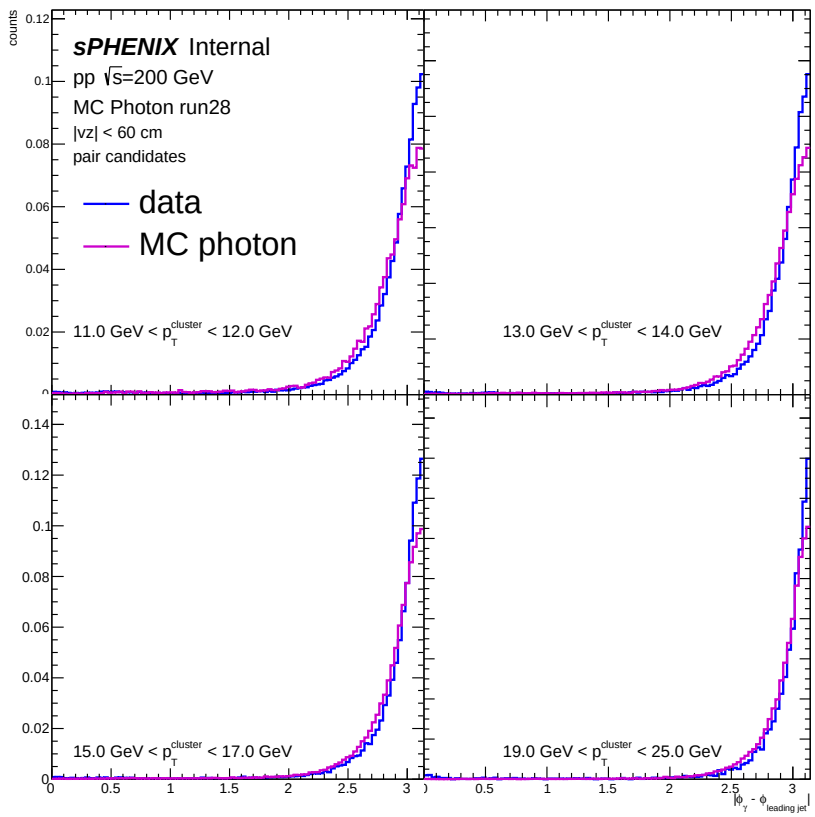

Figure 4: The $\Delta\phi$ between the EMCal cluster (photon candidate) and reconstructed R=0.4 jet for four example selections on EMCal cluster p_{T}. Shown are both data and Monte Carlo from the photon sample.

4 Analysis

The analysis steps are detailed in this section using the case of $R = 0.4$ jets, which is then extended to all other R values.

Once a pair has been determined, the photon is classified as a signal photon (region A) or an EMCal cluster in one of the three sideband regions (regions B, C, and D). The region determination is set by the cluster's isolation energy and BDT score, as shown in Figure 5. As a baseline, photon-jet pairs in region A are considered. To calculate the $\langle x_J \rangle$, the distributions are fit to a guassian and the arithmetic mean is taken, as in Figure 6.

The BDT score calulation is done using the `PhotonClusterBuilder` module found at: `https://github.com/sPHENIX-Collaboration/coresoftware/tree/master/offline/packages/CaloReco`

For the main body of this text, we will be using the arithmetic mean as the $\langle x_J \rangle$ extracted from these plots. The kinematic border is plotted as well for clarity, and when fitting the x_J distribution or taking the mean, only bins above the border are considered. The remaining distribution is fitted to a gaussian, and the fit's mean or the arithmetic mean is reported as that p_T bin's $\langle x_J \rangle$.

For both data and MC, the jets are JES corrected using the `JetCalib` module found at: `https://github.com/sPHENIX-Collaboration/coresoftware/tree/master/offline/packages/jetbase`.

Additionally, the JER blurring correction is added to the MC samples [6]

Finally, the mean of all p_T bins are collected and plotted together as $\langle x_J \rangle$ as a function of cluster p_T. There are insufficient statistics to determine a p_T dependent result and hence the fit is a simple one-parameter level. One can then check if this is consistent within statistics of no p_T dependence.

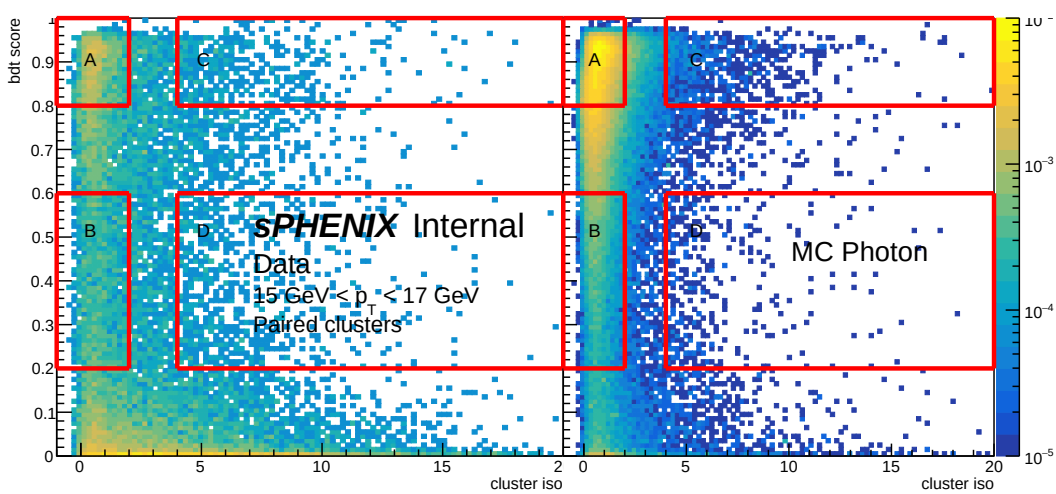

Figure 5: Isolation energy and BDT score of paired clusters. Clusters in the signal region, A, have an isolation energy < 2 GeV, and a BDT score > 0.8. Here the difference between data (left) and the MC Photon sample (right) are shown.

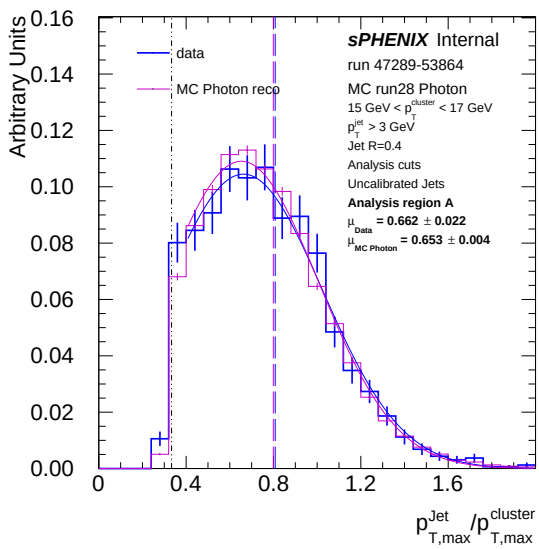

Figure 6: x_J for a single p_{T} bin with jets having $R = 0.4$. The black dotted line is drawn at the ratio of the minimum jet p_{T}/minimum cluster p_{T} and the colored dashed lines are drawn at the mean of the x_J distribution.

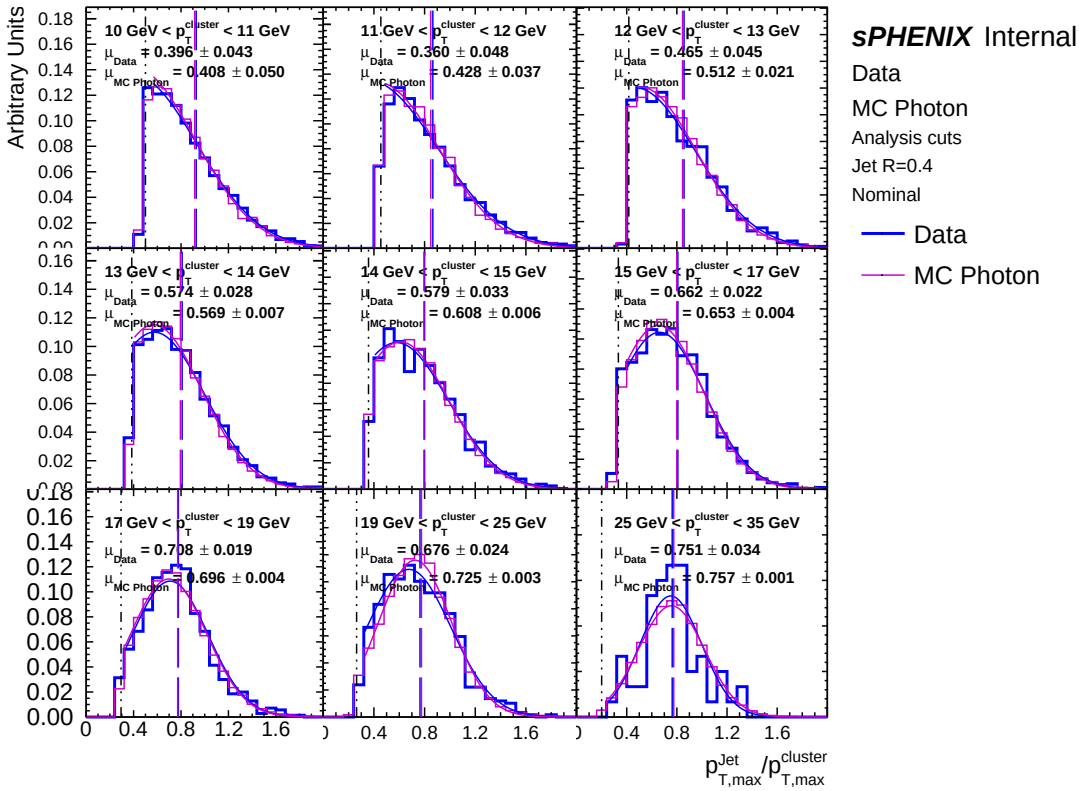

Figure 7: x_J for all p_T bins with jets having $R = 0.4$. The black dotted line is drawn at the ratio of the minimum jet p_T/minimum cluster p_T and the colored dashed lines show the mean of the distributions. Also listed for each p_T bin are the fitted mean values with uncertainties.

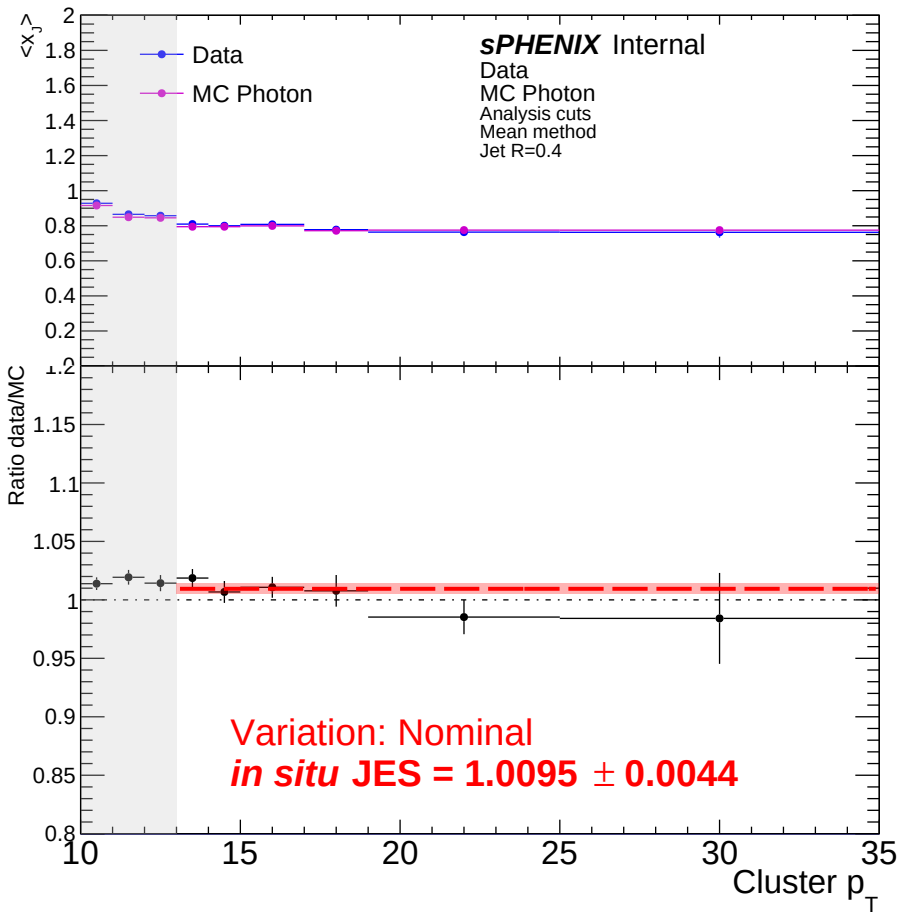

Figure 8: $\langle x_J \rangle$ as a function of cluster p_T for jets having $R = 0.4$. The horizontal line best fit is shown as the dashed curve (only fit for $p_T > 13$ GeV) and the level and uncertainty quoted.

5 Systematic Uncertainties

5.1 Photon Purity

The decision for the BDT score and isolation energy cut will still allow some signal leakage into the sideband regions, and background leakage into the signal region. To account for this uncertainty, we vary the BDT score cut and isolation energy separately in order to see the overall effect on the distributions.

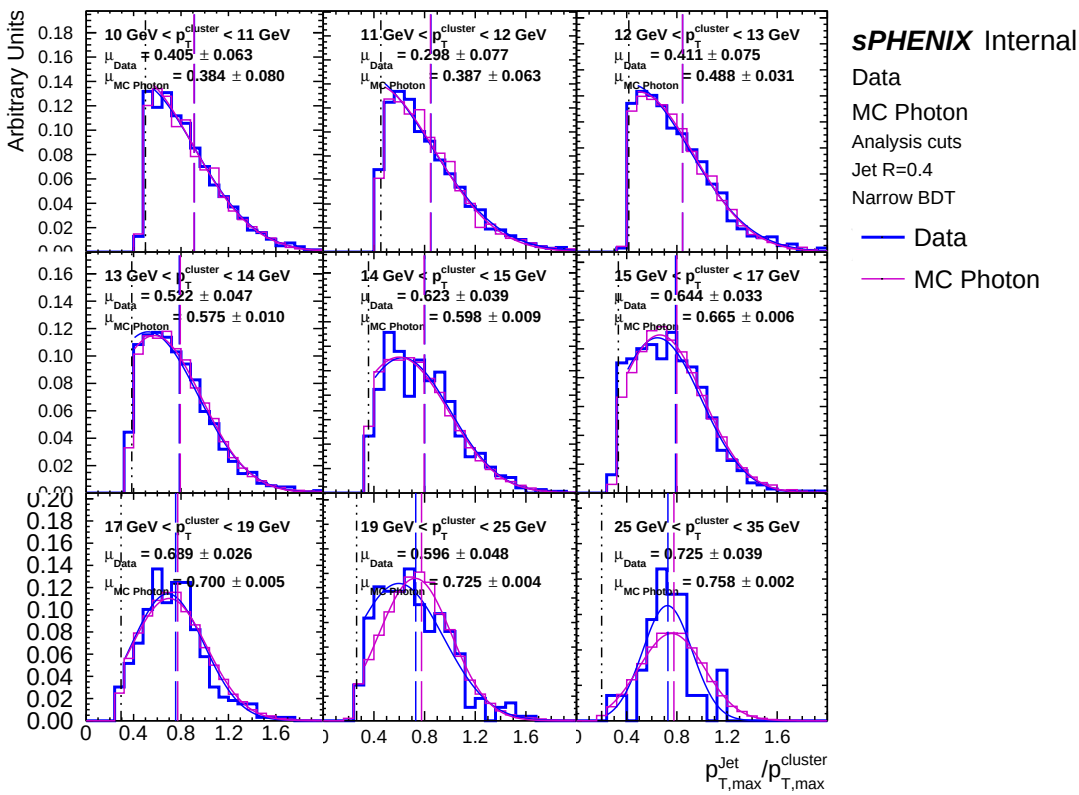

Figure 9: x_J for all p_T bins with jets having $R = 0.4$ and the narrower BDT requirement. The black dotted line is drawn at the ratio of the minimum jet p_T/minimum cluster p_T and the colored dashed lines show the mean of the distributions. Also listed are the fitted mean values with uncertainties

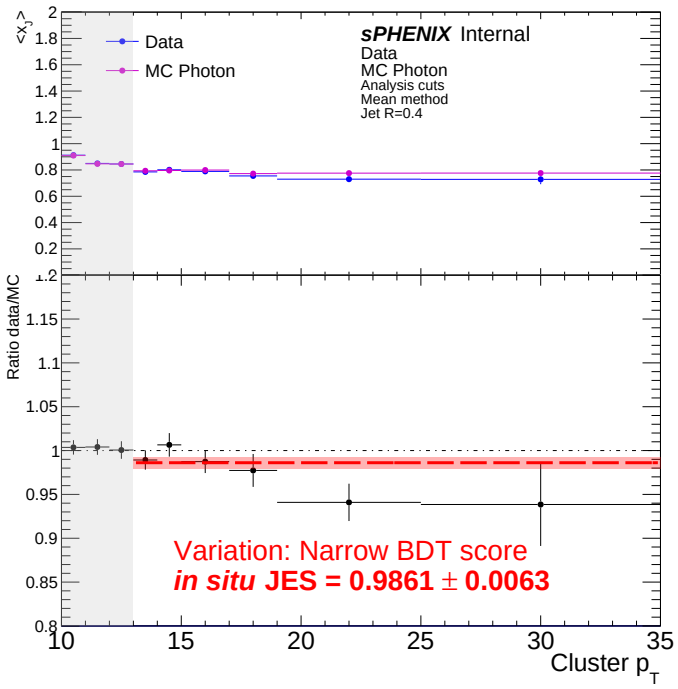

Figure 10: $\langle x_J \rangle$ is plotted as a function of cluster p_T for jets having $R = 0.4$ and applying the narrow BDT variation. The horizontal line best fit is shown as the dashed curve (only fit for p_T>13 GeV) and the level and uncertainty quoted.

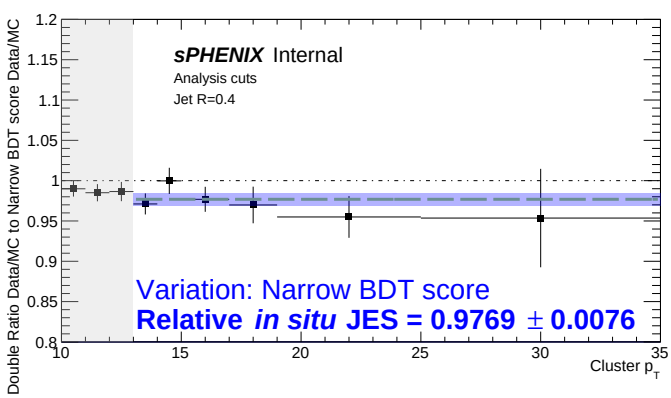

Figure 11: The double ratio of the nominal analysis's comparison of data to the Monte Carlo sample vs the narrow BDT variation's comparison of data to the Monte Carlo sample.

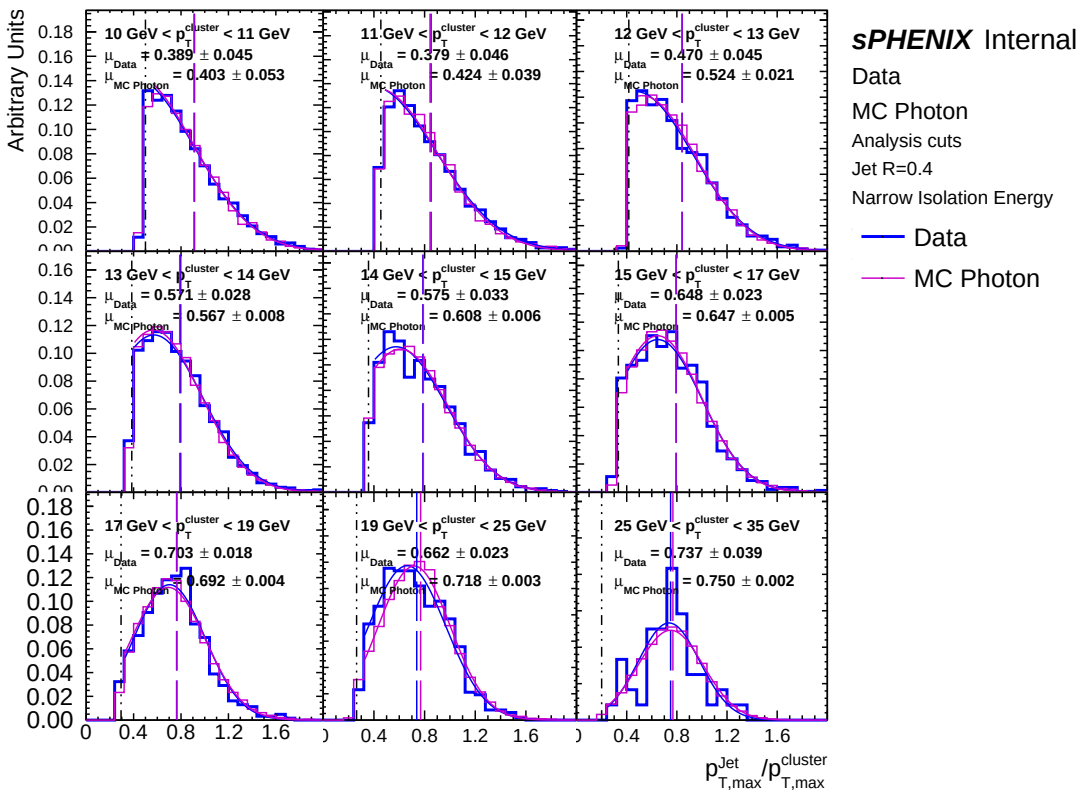

Figure 12: x_J for all p_T bins with jets having $R = 0.4$ and the narrower isolation energy requirement. The black dotted line is drawn at the ratio of the minimum jet p_T/minimum cluster p_T and the colored dashed lines show the mean of the distributions. Also listed are the fitted mean values with uncertainties

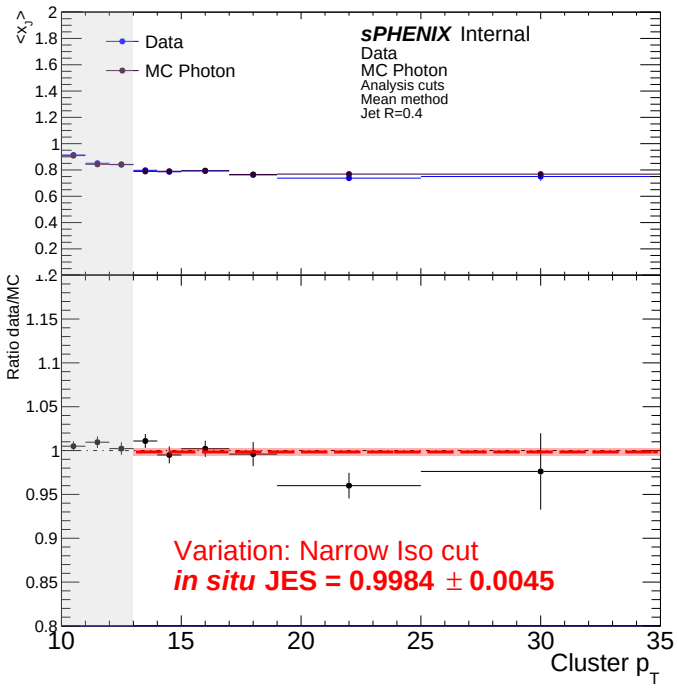

Figure 13: $\langle x_J \rangle$ is plotted as a function of cluster p_T for jets having $R = 0.4$ and applying the narrow isolation cut variation. The horizontal line best fit is shown as the dashed curve (only fit for p_T>13 GeV) and the level and uncertainty quoted.

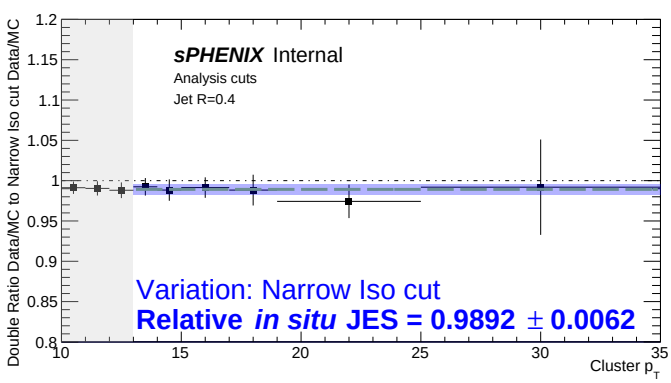

Figure 14: The double ratio of the nominal analysis's comparison of data to the Monte Carlo sample vs the narrow isolation energy cut variation's comparison of data to the Monte Carlo sample.

5.2 Jet Topology

Another variation considered for systematic uncertainty determination related to the event jet topology. In the standard γ-jet events, one can think of there being two "jets" - one from the photon and one from the opposite side jet. In this variation, one require the event to have no "third" jet to be present. One examines all jets in the event that are **not** the original two jets (requiring that the other jets are separated by $\Delta R > R_{jet}$). If there is any such "third" jet with $p_T^{jet} > 5$ GeV, the event is rejected. Notably, there is no pseudorapidity requirement on this "third" jet.

Figure 15 shows the probability of finding a "third" jet in the event as a function of Cluster p_{T} for both data and Monte Carlo. The data and Monte Carlo show qualitative agreement on the overall level and trend with p_{T}.

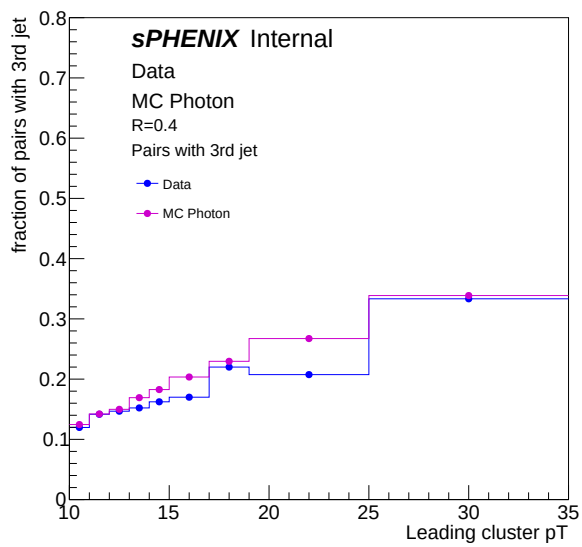

Figure 15: Shown is the probability that a γ-jet event has a "third" jet as defined in the text as a function of Cluster p_{T}. Results are shown from data and the Monte Carlo Photon sample.

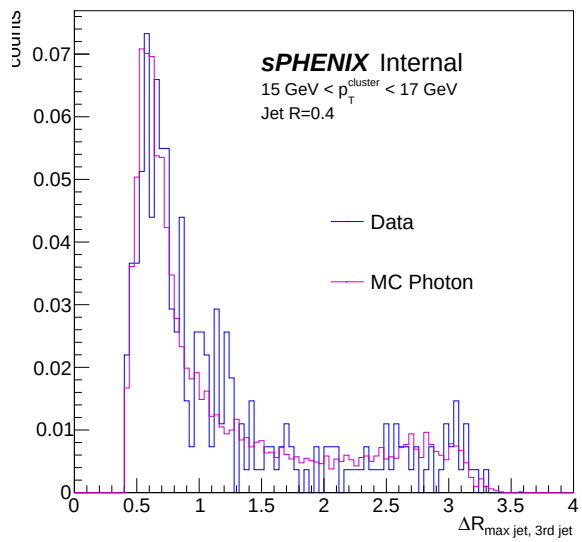

Figure 16: Shown is the ΔR between the the away-side jet and the "third" jet as defined in the text for a specific p_T range and jet radius. Results are shown from data and the Monte Carlo Photon sample.

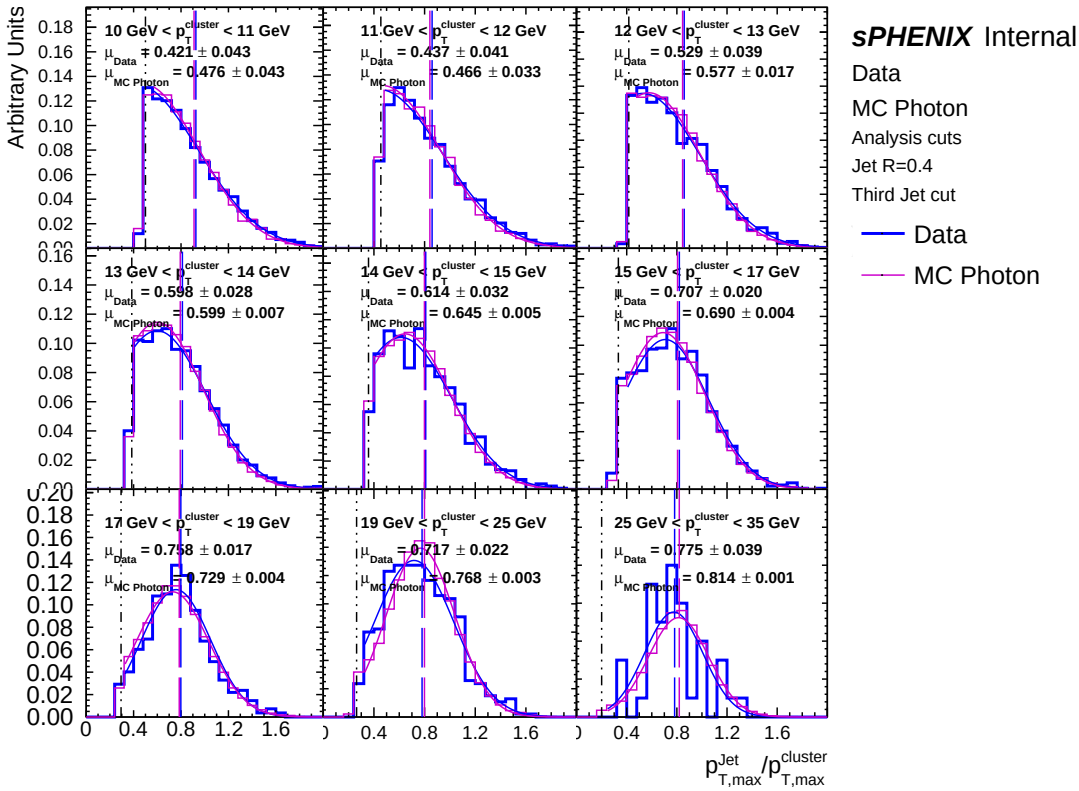

Figure 17: x_J for all p_T bins with jets having $R = 0.4$ and a cut on events with a third jet. The black dotted line is drawn at the ratio of the minimum jet p_T/minimum cluster p_T and the colored dashed lines show the mean of the distributions. Also listed are the fitted mean values with uncertainties

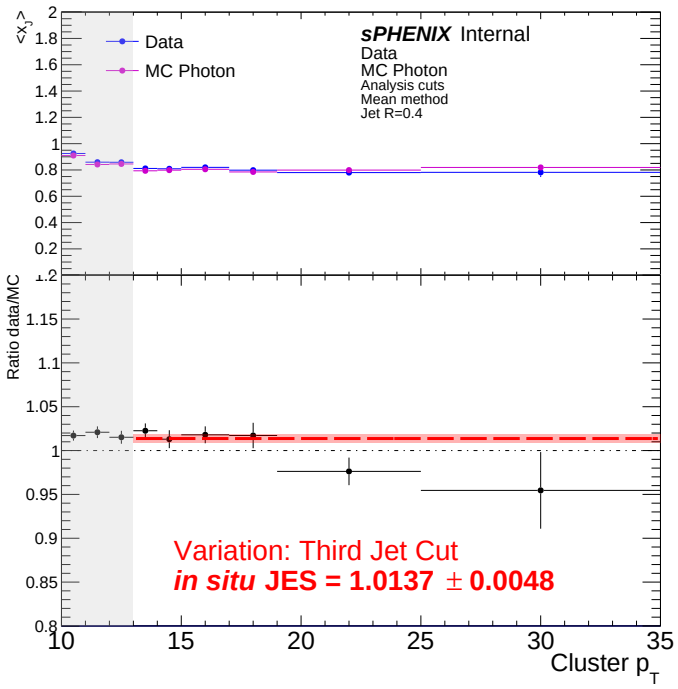

Figure 18: $\langle x_J \rangle$ is plotted as a function of cluster p_T for jets having $R = 0.4$ and applying the third jet cut variation. The horizontal line best fit is shown as the dashed curve (only fit for p_T>13 GeV) and the level and uncertainty quoted.

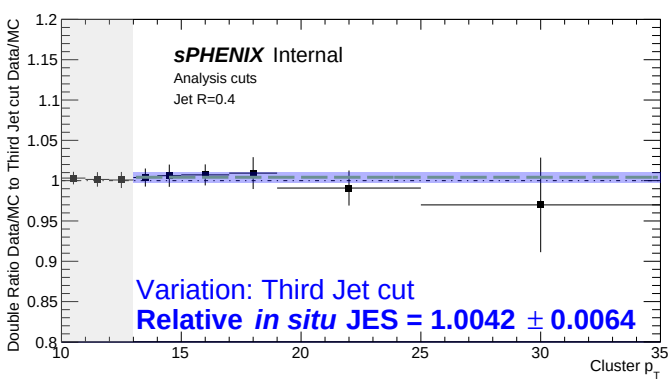

Figure 19: The double ratio of the nominal analysis's comparison of data to the Monte Carlo sample vs the third jet cut variation's comparison of data to the Monte Carlo sample.

5.3 Model Dependence

All of the comparisons made so far have been relative to PYTHIA-8, but other simulation models exist as well. HERWIG samples are being generated, and a comparison to that Monte Carlo data set will provide an additional systematic uncertainty contribution.

6 *In Situ* Results

Results for the *in situ* jet energy scale correction for different anti-k$_t$ jet sizes and their statistical and systematic uncertainties are provided in Table 1. Within statistical precision, there is no clear p_{T} dependence and so just these single values are quoted. It is also notable that within uncertainties, the scale correction is consistent for the R values examined.

Jet Radius	Nominal $x_{J\gamma}$	Purity	Isolation	Jet topology	Model
0.4	1.0095 $\pm$ 0.0044 (stat) $\pm$ 0.0259 (sys)	0.0231	0.0109	0.0043	-
0.6	0.9984 $\pm$ 0.0046 (stat) $\pm$ 0.0268 (sys)	0.0222	0.0136	0.0059	-
0.8	1.0012 $\pm$ 0.0058 (stat) $\pm$ 0.0291 (sys)	0.0250	0.0134	0.0063	-

Table 1: Table of *in situ* jet energy calibration results for different anti-k$_t$ jet size parameters.

The Model dependent systematic uncertainty will be detailed in Section 5.3. This is not yet complete as it awaits the HERWIG simulation sample. As soon as that is available, this will be made complete. It is possible that additional studies or corrections on the purity could be explored and might reduce that dominant source of systematic uncertainty.

7 Appendix: Photon and Jet Kinematic Checks

In this Appendix, we document the comparison of γ-jet kinematics in data and the Monte Carlo Photon sample. Figure 20 shows the p_T distribution of EMCal clusters passing all photon cuts documented in this note from data and the Monte Carlo Photon sample. The bottom panel shows the ratio of MC to data. There is a difference in the p_T slope, and currently no re-weighting of the MC is done in the default analysis Since the analysis is done in relatively narrow bins in p_T, this impact should be very small. Figures 21 and 22 shows the pseudorapidity distribution (left) and $\eta - \phi$ maps (right) for the photons and partner jets, respectively. Both show reasonable agreement.

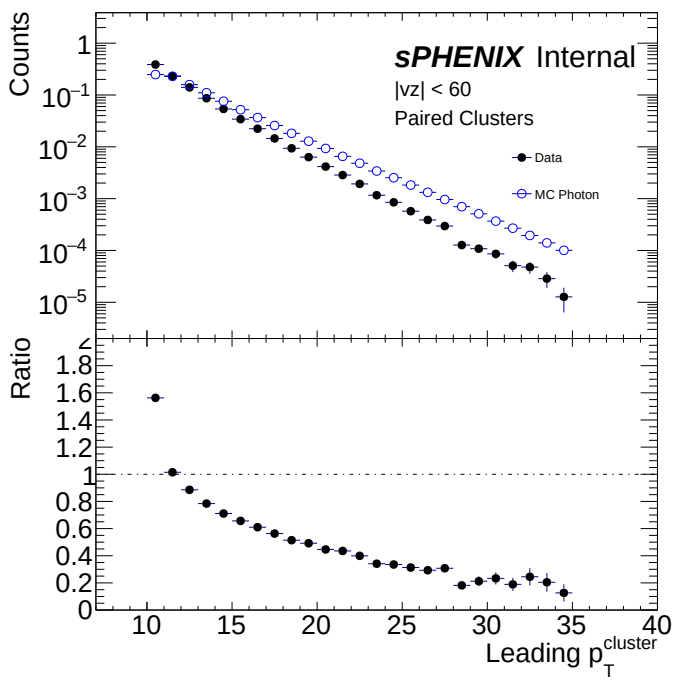

Figure 20: Data vs MC cluster p_T in the analysis p_T range.

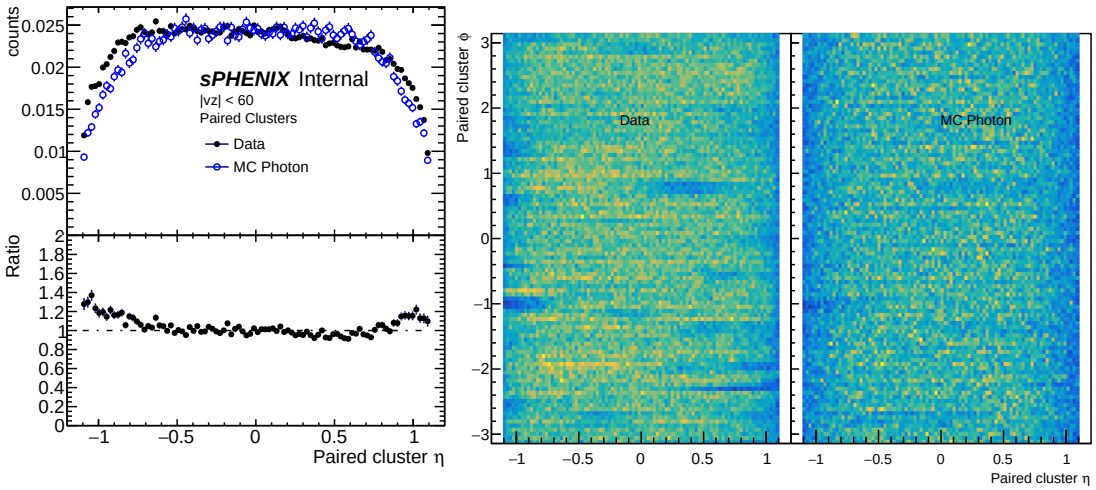

Figure 21: Data vs MC cluster η and ϕ distributions for paired clusters

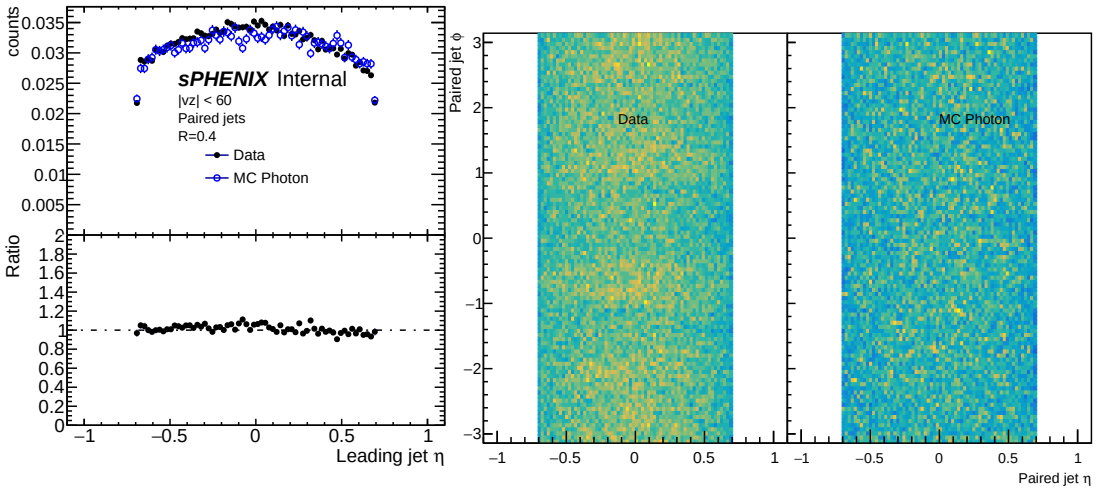

Figure 22: Data vs MC jet η and ϕ distributions for paired jets with R=0.4

8 Appendix: Alternative Gaussian Fitting Method

An alternative method for characterizing the x_J distribution was explored, i.e., fitting the x_J distribution to a Gaussian function. In that case, the distribution in data and Monte Carlo are compared by the ratio of the Gaussian mean values. This is in contrast to the default method detailed in the main note using the arithmetic mean of the x_J distribution. This section shows the results using the Gaussian fit means again for the R=0.4 jet case. Final results for this method are seen in 2

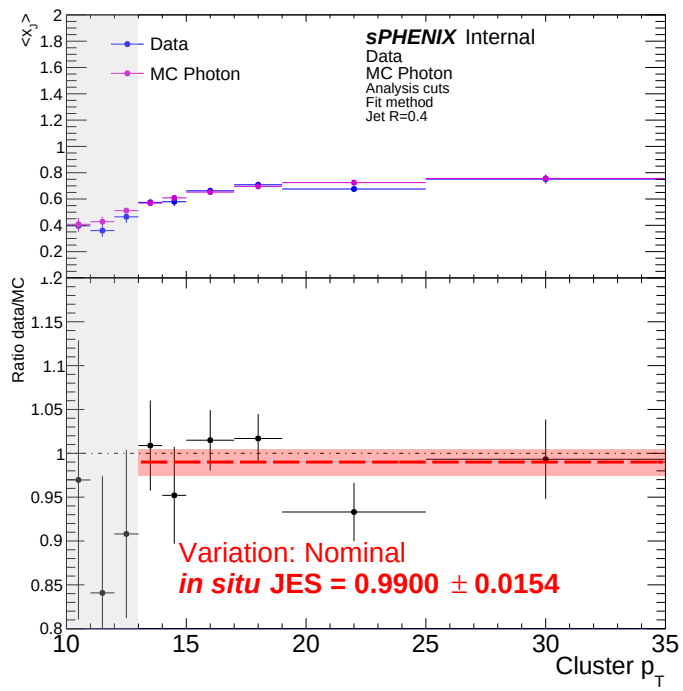

Figure 23: $\langle x_J \rangle$ as found by fitting the distributions is plotted as a function of cluster p_T for jets having $R = 0.4$. The horizontal line best fit is shown as the dashed curve (only fit for p_T>13 GeV) and the level and uncertainty quoted.

Jet Radius	Nominal $x_{J\gamma}$	Purity	Isolation	Jet topology	Model
0.4	0.9900 ± 0.0154 (stat) ± 0.0356 (sys)	0.0344	0.0072	0.0056	-
0.6	0.9973 ± 0.0086 (stat) ± 0.0268 (sys)	0.0246	0.0102	0.0029	-
0.8	1.0025 ± 0.0089 (stat) ± 0.0347 (sys)	0.0304	0.0120	0.0116	-

Table 2: Table of *in situ* jet energy calibration results for different anti-k_t jet size parameters found by using a Gaussian fit instead of the arithmetic mean on the x_J distributions.

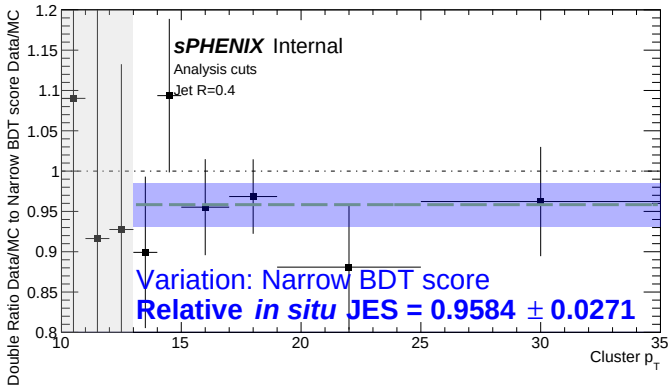

Figure 24: The double ratio of the comparison of the nominal analysis using the fit method of calculating $\langle x_J \rangle$ of data to the Monte Carlo sample vs the narrow BDT variation's comparison of data to the Monte Carlo sample.

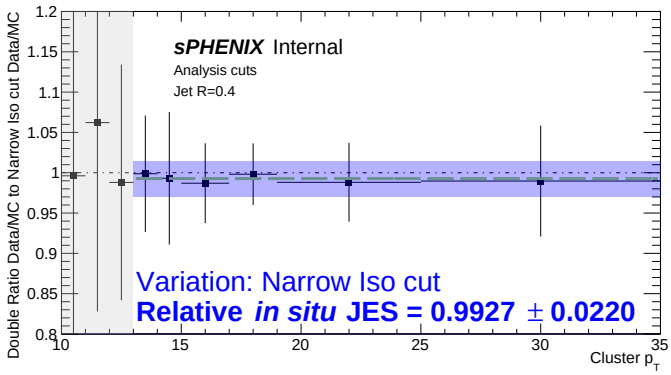

Figure 25: The double ratio of the comparison of the nominal analysis using the fit method of calculating $\langle x_J \rangle$ of data to the Monte Carlo sample vs the narrow isolation energy variation's comparison of data to the Monte Carlo sample.

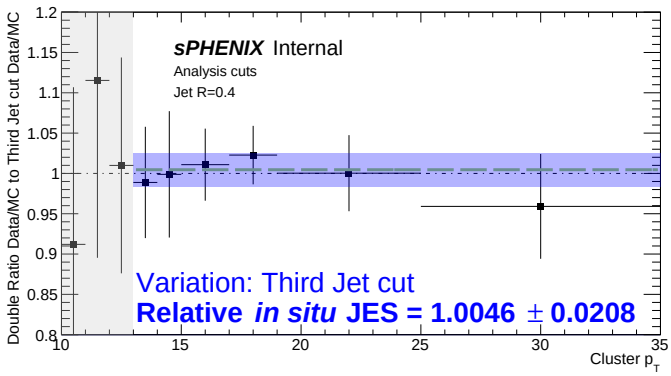

Figure 26: The double ratio of the comparison of the nominal analysis using the fit method of calculating $\langle x_J \rangle$ of data to the Monte Carlo sample vs the third jet cut variation's comparison of data to the Monte Carlo sample.

9 Appendix: R=0.6 Jet Figures

The main text of this note focuses on R=0.4 jets as a default, but the same set of plots exist for R=0.6, and R=0.8 jets. Here we show the main set of analysis plots using R=0.6 jets. The final results are already listed in the table above.

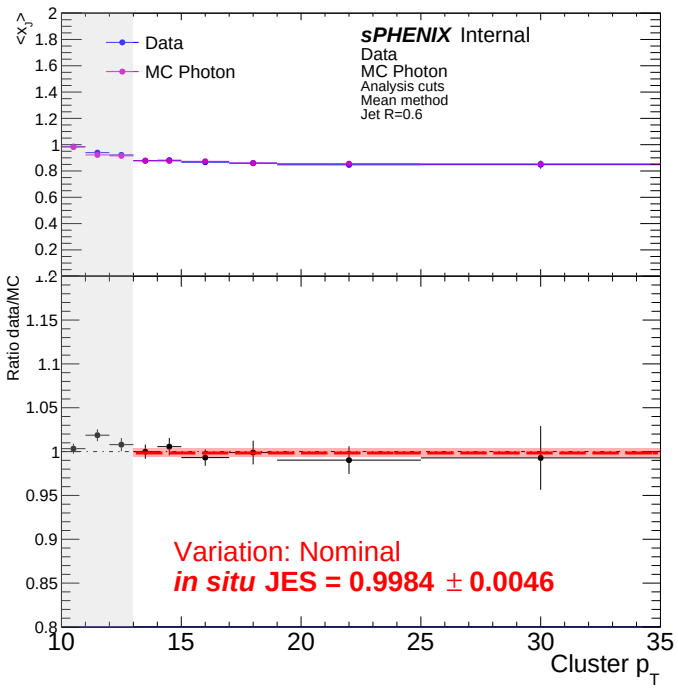

Figure 27: $\langle x_J \rangle$ is plotted as a function of cluster p_T for jets having $R = 0.6$. The horizontal line best fit is shown as the dashed curve (only fit for p_T>13 GeV) and the level and uncertainty quoted.

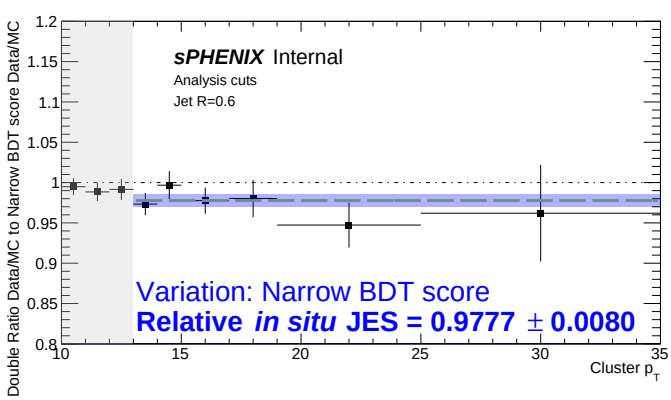

Figure 28: The double ratio of the nominal analysis's with $R = 0.6$ jets comparison of data to the Monte Carlo sample vs the narrow BDT score variation's comparison of data to the Monte Carlo sample.

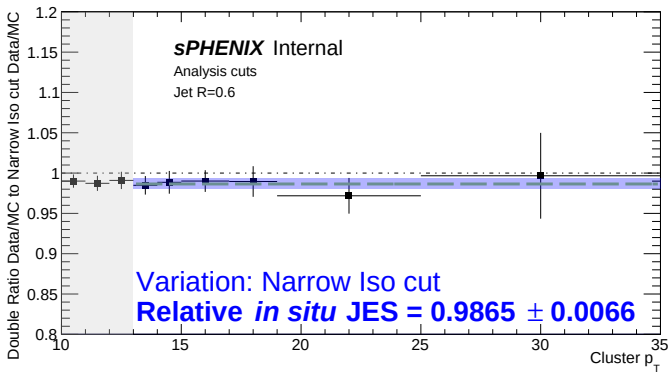

Figure 29: The double ratio of the nominal analysis's with $R = 0.6$ jets comparison of data to the Monte Carlo sample vs the narrow isolation energy cut variation's comparison of data to the Monte Carlo sample.

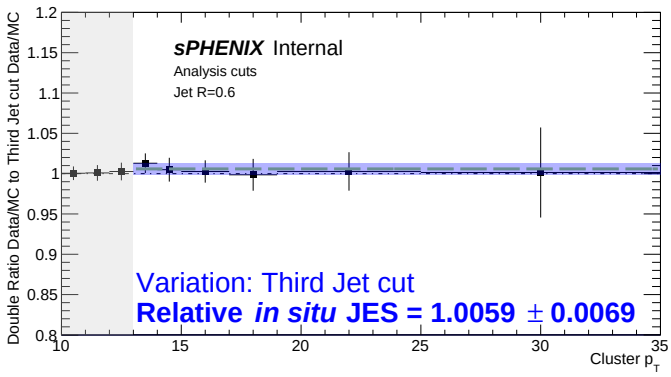

Figure 30: The double ratio of the nominal analysis's with $R = 0.6$ jets comparison of data to the Monte Carlo sample vs the third jet cut variation's comparison of data to the Monte Carlo sample.

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